Ananya Guduri

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LinkedIn



Education

Manipal Institute of Technology — Manipal, India
B.Tech. in Electronics and Communication Engineering — *Expected 2028*

CGPA: 8.86/10.00
Relevant Coursework: Digital System Design, Verilog, Analog Electronics, VLSI

Technical Skills

Languages: Verilog HDL, C
Tools: Vivado, Cadence Virtuoso, LTspice, KiCad, Proteus
Foundations: Digital logic, RTL design, simulation testbenches, analog circuit simulation
Learning Focus: SystemVerilog, functional verification, static timing analysis

Projects & Research

Dual-Output XOR–XNOR Logic: Design and Power Analysis *Research*

- Co-authored research on a dual-output XOR–XNOR circuit using static CMOS and pass-transistor logic, simulated in Cadence Virtuoso.
- Demonstrated reductions in static and dynamic power compared with separate CMOS XOR and XNOR implementations through DC and transient simulations at a 1.8 V supply.

Analog Extravasation Detection Prototype *Hardware*

- Developed an analog hardware proof of concept to detect IV fluid extravasation through electrical signal variations, using analog circuitry without a microcontroller or digital processing.

Experience

Aerionix 2025
Electronics Intern

- Contributed to electronic speed controller (ESC) development at a student-founded startup working on BLDC motors and flight-control electronics, gaining practical circuit-design and testing experience.

Robomanipal — Manipal, India
Electronics Taskphase Member — *Sep 2024 – Mar 2025*

- Completed introductory circuit-design and simulation tasks using LTspice, Proteus and KiCad, covering schematic capture, analog and digital circuits, and PCB layout workflows.

Leadership & Technical Activities

Vice President; Simulation & Design Head, ChipMIT

- Leading technical workshops and collaborative projects; mentoring junior members in Verilog, RTL debugging, and analog circuit simulation.

Additional Roles: Electronics Domain Head, Management Committee, IE E&C; Management Committee, IEEE Student Body Manipal; Organizing Committee, QUARK, TechTatva 2025.

Achievement

Second Place, Ecocircuit – TechTatva 2024, Manipal Institute of Technology.